

ANNE CHRISTIONO

anchris@mit.edu | +1 (281) 725-8561 | Cambridge, MA, USA | [anchris24.github.io/](https://github.com/anchris24)

EDUCATION:

Massachusetts Institute of Technology, Cambridge, MA

August 2024 - May 2027

Bachelor of Science in Computer Science & Engineering and Mathematics

GPA: 5.0/5.0

Courses: Deep Learning, Machine Learning, Algorithms, Statistics, Probability & Random Variables, Linear Algebra, Programming
Activities: Machine Learning L.A., Multivariable Calc Grader, HMMT Staff, MIT Hacking Medicine, Society of Women Engineers

SKILLS:

Languages & Tools: Python, Java, Pytorch, Scikit-learn, SHAP, SQL, Postgres, C++, JavaScript, HTML/CSS, LangChain

AI/ML Concepts: Feature Selection, Model Explainability, Data Visualization, Sentiment Analysis, Data Structures, Causal Graphs

PROFESSIONAL EXPERIENCE:

Microsystems Technology Laboratories

Cambridge, MA, USA

Full-Stack Software Engineering Intern

Mar 2025 – present

- Integrating transformer-based LLMs with PostgreSQL for semantic search, query parsing, & automated patent summarization
- Developing full-stack system with Python backend, REST APIs, and HTML/CSS/JS frontend for scalable patent discovery
- Engineering data pipelines and APIs to process dataset of 100+ patents with relational mappings between 50 lab members

National University of Singapore (NUS) Computing: AI for Social Good Lab

Singapore

Human-AI Interaction Undergraduate Researcher

Mar 2025 – Aug 2025

- Built multimodal AI integrating LLMs with TTS/STT, enabling real-time sentiment detection in human-bot interviews
- Extracted text and voice features, applying ML-driven causal graph analysis to reveal key factors behind mental health stigma

Houston Methodist Research Institute: Mathematics in Medicine Department

Houston, TX, USA

Machine Learning Research Intern

May 2023 – Jun 2025

- Engineered and evaluated 18 tree-based, nonlinear, and linear predictive models (XGBoost, CatBoost, SVM, KNN) with Scikit-learn to classify nanoparticle toxicity using curated dataset of 8,000+ samples and 19 physicochemical features
- Implemented nested 10-fold cross-validation and GridSearch for hyperparameter optimization and robust model selection
- Utilized SHAP analysis for model explainability & feature selection; quantified generalizability (88.7% accuracy on new data)

MIT Media Lab: Sculpting Evolution Lab

Cambridge, MA, USA

AI/ML Undergraduate Researcher

Sep 2024 – Jan 2025

- Developed PyLabRobot, versatile cross-platform Python interface to program wide range of liquid-handling Opentrons robots
- Leveraged LangChain (LLMs) to implement PyLabRobot features to execute experiments based on natural language prompts

The MITRE Corporation

McLean, VA, USA

Data Science Research Intern

Jun 2022 – Aug 2022

- Applied statistical modeling, data analysis, and market research on iGEM datasets, literature, and SME interviews in SynBio & Computational Biology for DoD-sponsored SynBio Moonshot, generating data-driven insights to guide R&D strategies
- Published internal reports and presented to MITRE leadership, influencing strategic direction in synthetic biology innovation

PUBLICATIONS:

- Cave, J., **Christiono, A.**, et al. (2025). Rational Design of Safer Inorganic Nanoparticles Via Mechanistic Modeling-informed Machine Learning. *ACS Nano*, 19(23), 21538–21555. <https://doi.org/10.1021/acsnano.5c03590>
- Christiono, A.**, et al. (2022). The Wiener Index: From Trees to Graphs with Many Cut-Edges [Conference presentation abstract]. 53rd Southeastern Int'l Conference in Combinatorics, Graph Theory, and Computing, Florida Atlantic University.

LEADERSHIP AND COMMUNITY INITIATIVES:

MIT Department of EECS: Introduction to Machine Learning (6.390) Lab Assistant

Aug 2025 – present

- Develop lab questions reinforcing core ML concepts and encouraging broader consideration of applied ML in social contexts
- Hold office hours to guide students through debugging, model design, and interpretation of ML results with conceptual clarity

NUS High School of Math and Science: Teacher for Advanced Calculus & Machine Learning

Jan 2025 – Feb 2025

- Prepared and led calculus and ML courses for penultimate-year students as part of MIT's Global Teaching Labs in Singapore
- Designed slides, assessments, & interactive activities to promote discussion, analytical reasoning, & applied problem-solving

INTEGIRLS Houston: Advisor (2024-), Chapter Director (2022-2024), Web Director (2021-2022)

Aug 2021 – Jul 2024

- Global youth-led nonprofit to bridge gender gap in STEM through free, virtual math contests for female & nonbinary students
- Led 9 team members in outreach, fundraising, & graphic design to organize contests (participants doubled to 130+ globally)
- Raised \$10K+ from sponsors annually (doubled from prior year, newly acquired AOPS, HRT, Jane Street, Geogebra sponsors)

HONORS AND AWARDS:

- USA Computing Olympiad (USACO): Gold Division Competitor, 2021
- National Center for Women & Information Technology: 2x National Honorable Mention & Houston Award Winner, 2022-23
- American Invitational Mathematics Examinations (AIME): 5x Qualifier, 2020-2024